Class 11: AlphaFold

Aadhya Tripathi (PID: A17878439)

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Background

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The PDB database (the main repository of experimental structures) only has ~**250 thousand** structures, as we saw in the last lab. The main protein sequence database has **>200 million** sequences! Only 0.125% of known sequences have a known structure. This is called the "structure knowledge gap".

```
250000 / 200000000
```

```
[1] 0.00125
```

- Structures are much harder to determine than sequences.
- They are expensive (costing $1M on average).
- They take an average of 3-5 years to solve.

EBI AlphaFold Database

The EBI has a database of pre-computed AlphaFold (AF) models called AFDB. This is growing all the time and can be useful to check before running AF ourselves.

Running AlphaFold

We can download and run locally but we need a GPU. Or we can use "cloud" computing to run this on someone else's computer!

We will use ColabFold https://github.com/sokrypton/ColabFold

We previously found there was no AFDB entry for our HIV sequence.

```
>HIV-Pr-Dimer
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQ
```

Here we will use AlphaFold2_mmseqs2.

Figure 1: Mol* Visualization of Model 1, colored by pLDDT confidence

Analysis of Resulting Models

Save AlphaFold2 prediction results and import

```
results_dir <- "HIV_23119/"
```

```
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

# Print our PDB file names
basename(pdb_files)
```

```
[1] "HIV_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIV_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIV_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIV_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIV_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

Load bio3d and save top 5 ranked models

```
library(bio3d)
```

```
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIV_23119/HIV_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
HIV_23119/HIV_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
HIV_23119/HIV_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
HIV_23119/HIV_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
HIV_23119/HIV_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: HIV_23119/HIV_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed
pdb/seq: 2   name: HIV_23119/HIV_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed
pdb/seq: 3   name: HIV_23119/HIV_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed
pdb/seq: 4   name: HIV_23119/HIV_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed
pdb/seq: 5   name: HIV_23119/HIV_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed
```

```
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]HIV_23119_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIV_23119_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIV_23119_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIV_23119_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIV_23119_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]HIV_23119_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]HIV_23119_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIV_23119_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIV_23119_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIV_23119_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .         100

                            101        .         .         .         .         150
[Truncated_Name:1]HIV_23119_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]HIV_23119_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]HIV_23119_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]HIV_23119_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]HIV_23119_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                              **************************************************
                            101        .         .         .         .         150

                            151        .         .         .         .       198
[Truncated_Name:1]HIV_23119_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIV_23119_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIV_23119_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIV_23119_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIV_23119_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                              **************************************************
                            151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
```

```
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Calculate RMSD between all pairs models:

```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

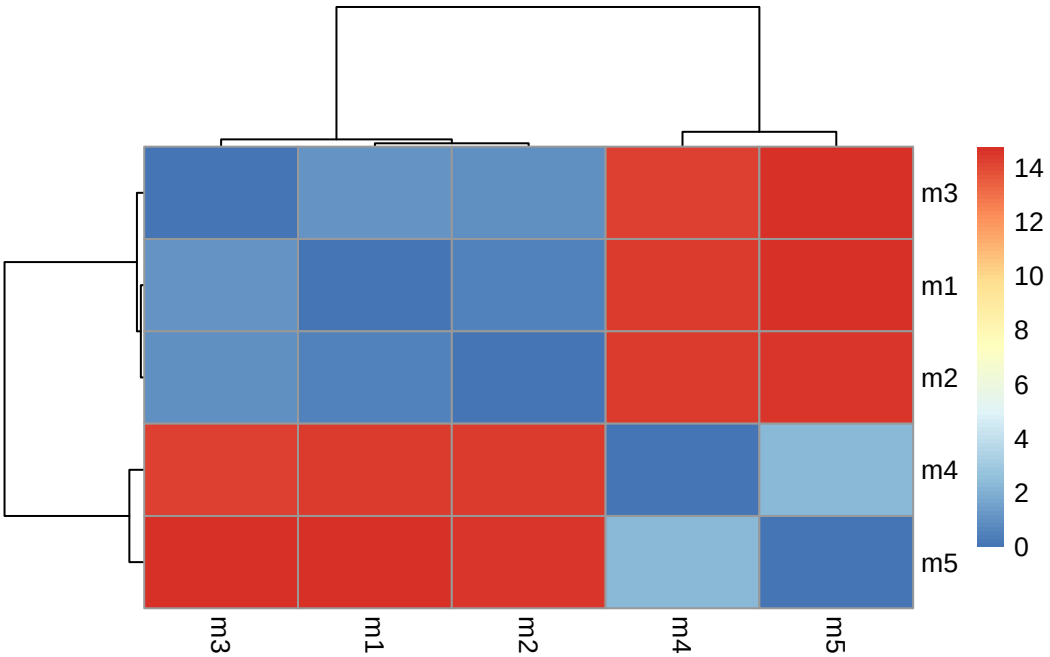
```r
range(rd)
```

```
[1]  0.000 14.754
```

```r
library(pheatmap)
```

Heatmap of pairs models:

```r
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```

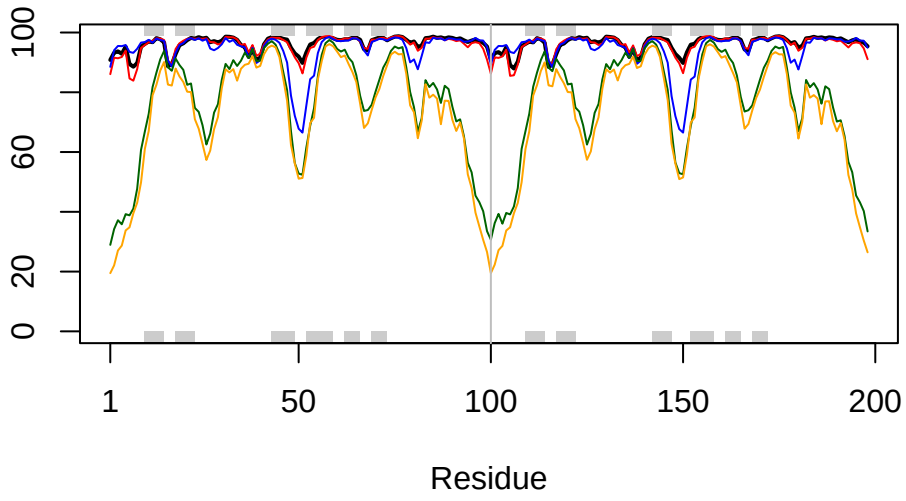



```
# reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

Plot the pLDDT values across all models:

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```


100
60
20
0
1
50
100
150
200
Residue

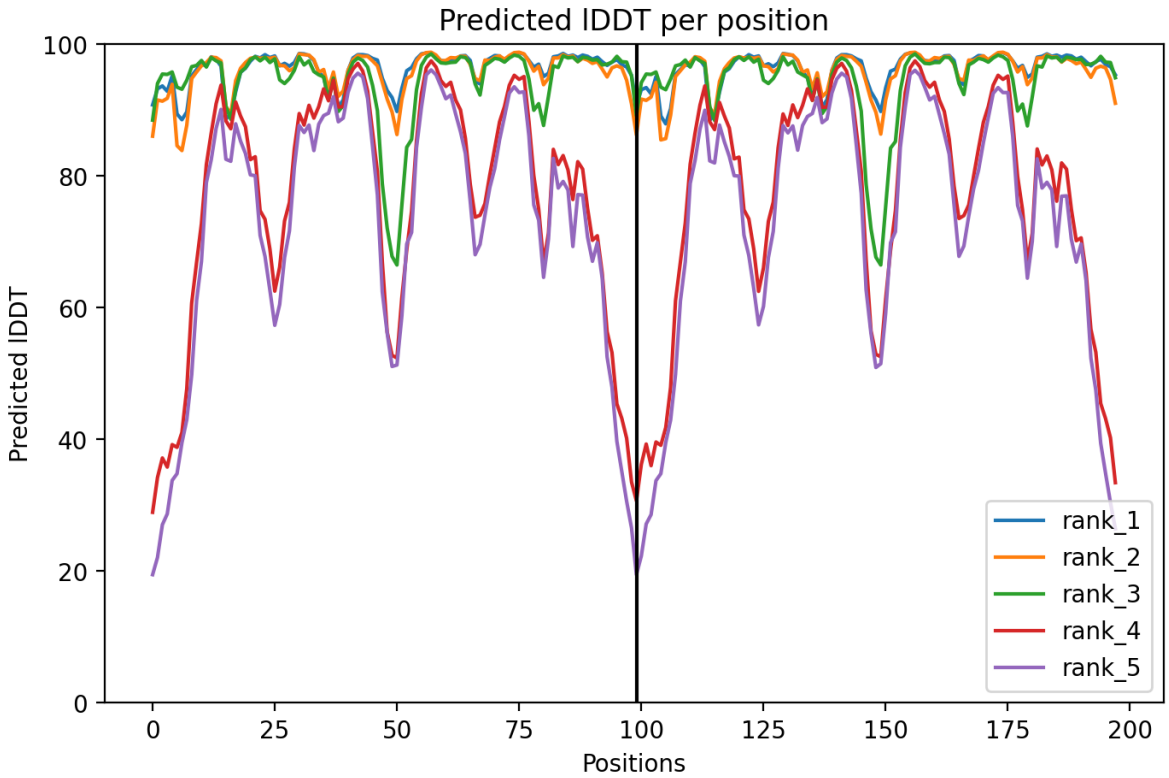


Figure 2: pLDDT plot from AlphaFold

Find most consistent "rigid core" common across all the models:

```
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9885.822
core size 196 of 198  vol = 6896.71
core size 195 of 198  vol = 1337.847
core size 194 of 198  vol = 1040.67
core size 193 of 198  vol = 951.857
core size 192 of 198  vol = 899.083
core size 191 of 198  vol = 834.732
core size 190 of 198  vol = 771.338
core size 189 of 198  vol = 733.065
core size 188 of 198  vol = 697.28
core size 187 of 198  vol = 659.742
core size 186 of 198  vol = 625.273
core size 185 of 198  vol = 589.541
```

```
core size 184 of 198  vol = 568.253
core size 183 of 198  vol = 545.015
core size 182 of 198  vol = 512.889
core size 181 of 198  vol = 490.723
core size 180 of 198  vol = 470.266
core size 179 of 198  vol = 450.731
core size 178 of 198  vol = 434.735
core size 177 of 198  vol = 420.337
core size 176 of 198  vol = 406.658
core size 175 of 198  vol = 393.334
core size 174 of 198  vol = 382.395
core size 173 of 198  vol = 372.858
core size 172 of 198  vol = 356.994
core size 171 of 198  vol = 346.567
core size 170 of 198  vol = 337.446
core size 169 of 198  vol = 326.659
core size 168 of 198  vol = 314.95
core size 167 of 198  vol = 304.127
core size 166 of 198  vol = 294.552
core size 165 of 198  vol = 285.648
core size 164 of 198  vol = 278.884
core size 163 of 198  vol = 266.765
core size 162 of 198  vol = 258.994
core size 161 of 198  vol = 247.723
core size 160 of 198  vol = 239.84
core size 159 of 198  vol = 234.963
core size 158 of 198  vol = 230.062
core size 157 of 198  vol = 221.985
core size 156 of 198  vol = 215.62
core size 155 of 198  vol = 206.793
core size 154 of 198  vol = 196.984
core size 153 of 198  vol = 188.539
core size 152 of 198  vol = 182.262
core size 151 of 198  vol = 176.954
core size 150 of 198  vol = 170.712
core size 149 of 198  vol = 166.119
core size 148 of 198  vol = 159.796
core size 147 of 198  vol = 153.767
core size 146 of 198  vol = 149.092
core size 145 of 198  vol = 143.657
core size 144 of 198  vol = 137.138
core size 143 of 198  vol = 132.517
core size 142 of 198  vol = 127.231
```

```
core size 141 of 198  vol = 121.574
core size 140 of 198  vol = 116.775
core size 139 of 198  vol = 112.57
core size 138 of 198  vol = 108.17
core size 137 of 198  vol = 105.133
core size 136 of 198  vol = 101.249
core size 135 of 198  vol = 97.374
core size 134 of 198  vol = 92.974
core size 133 of 198  vol = 88.184
core size 132 of 198  vol = 84.029
core size 131 of 198  vol = 81.898
core size 130 of 198  vol = 78.019
core size 129 of 198  vol = 75.272
core size 128 of 198  vol = 73.052
core size 127 of 198  vol = 70.695
core size 126 of 198  vol = 68.975
core size 125 of 198  vol = 66.694
core size 124 of 198  vol = 64.394
core size 123 of 198  vol = 62.092
core size 122 of 198  vol = 59.045
core size 121 of 198  vol = 56.629
core size 120 of 198  vol = 54.016
core size 119 of 198  vol = 51.806
core size 118 of 198  vol = 49.652
core size 117 of 198  vol = 48.193
core size 116 of 198  vol = 46.648
core size 115 of 198  vol = 44.752
core size 114 of 198  vol = 43.292
core size 113 of 198  vol = 41.093
core size 112 of 198  vol = 39.147
core size 111 of 198  vol = 36.472
core size 110 of 198  vol = 34.117
core size 109 of 198  vol = 31.47
core size 108 of 198  vol = 29.448
core size 107 of 198  vol = 27.325
core size 106 of 198  vol = 25.822
core size 105 of 198  vol = 24.15
core size 104 of 198  vol = 22.648
core size 103 of 198  vol = 21.069
core size 102 of 198  vol = 19.953
core size 101 of 198  vol = 18.3
core size 100 of 198  vol = 15.723
core size 99 of 198  vol = 14.841
```

```
 core size 98 of 198  vol = 11.646
 core size 97 of 198  vol = 9.434
 core size 96 of 198  vol = 7.354
 core size 95 of 198  vol = 6.179
 core size 94 of 198  vol = 5.666
 core size 93 of 198  vol = 4.705
 core size 92 of 198  vol = 3.665
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.151
 core size 89 of 198  vol = 1.715
 core size 88 of 198  vol = 1.15
 core size 87 of 198  vol = 0.874
 core size 86 of 198  vol = 0.685
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

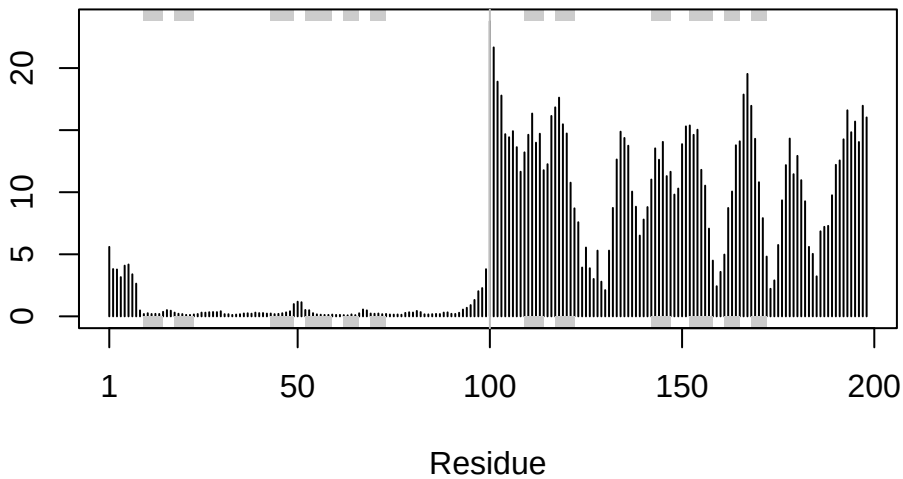
```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

RMSF between positions of the structure:

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

```
library(jsonlite)
```

Access Predicted Aligned Error files:

```
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

PAE files for Model 1 and 5:

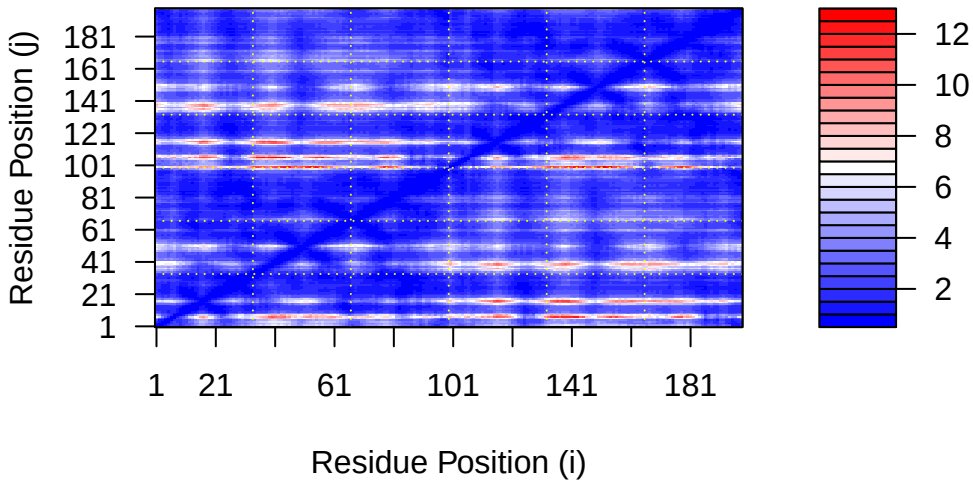
```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

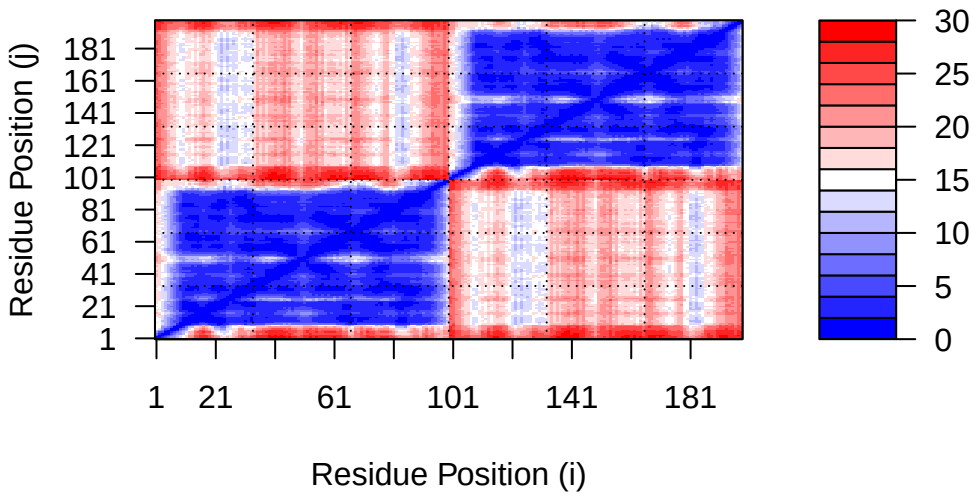
Plot N by N PAE scores for Model 1:

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```



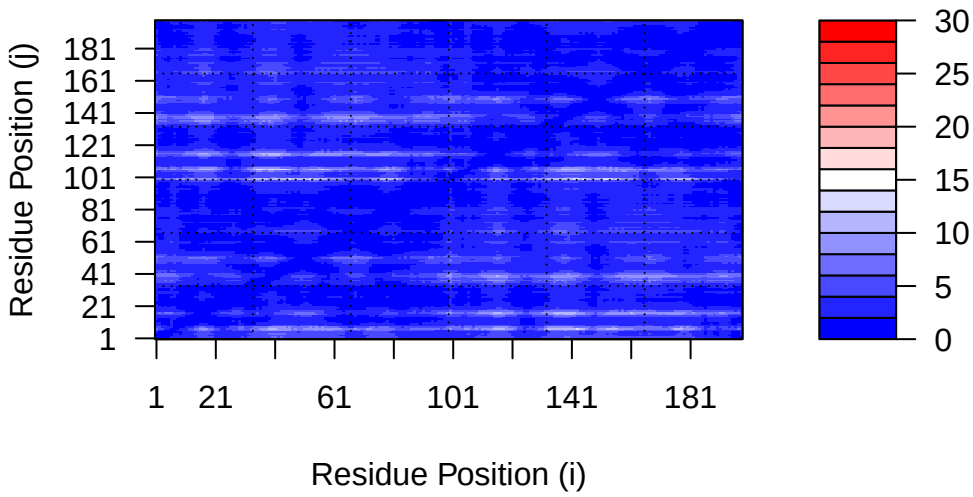
Plot N by N PAE scores for Model 5:

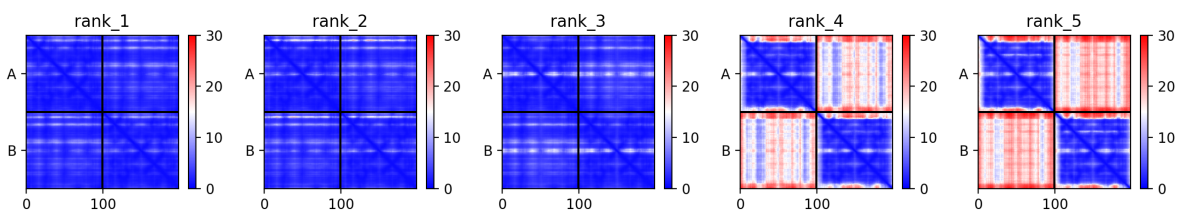
```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Plot N by N PAE scores for Model 1 again, using the same data range as the plot for model 5:

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```





Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIV_23119/HIV_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

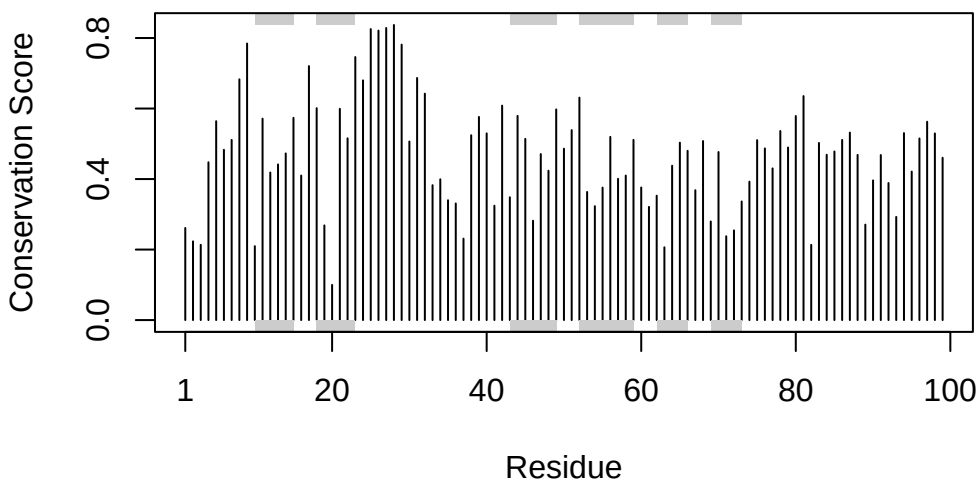
Number of sequences in alignment

```r
dim(aln$ali)
```

```
[1] 5397  132
```

Score residue conservation in the alignment

```r
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Conserved active sites:

```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

Save a PDB file with an Occupancy column with conservation scores for visualization.

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```